

Landau-Lifshitz equation with nonlocal damping due to exciton bath in two-dimensional magnetic semiconductors

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CrSBr, a magnetic van der Waals semiconductor, has shown exceptional coupling between their exciton and magnon degrees of freedom. Recently, it has been shown that light-driven exciton in CrSBr can also exert torque on localized magnetic moments residing in Cr atoms. Thereby, exciton order can be used not only to passively visualize magnon dynamics but also to actively manipulate them. Here, we consider a minimal microscopic model of CrSBr which unravels...

Introduction—The Landau-Lifshitz equation [1] with Gilbert [2] (LLG) form of damping has been the cornerstone of modeling basic research experiments in spintronics and magnonics, as well as applications in the magnetic recothe magnetic industry, usually via atomistic spin dynamics [3] or micromagnetics [4, 5] software packages. However, the LLG equation is phenomenological in nature, so its applicability must be tested in every new type of experimental situation [6] which often requires additional terms fitted to experiments, as a local in space and time Gilbert damping term parametrized by a single constant cannot capture the complexity of phenomena encountered [7]. One novel phenomenon observed recently due to the arrival of two-dimensional magnetic semiconductors is femtosecond laser pulse (fsLP) excitation of magnons via exciton mediation. For example, this has been reported for 2D antiferromagnetic semiconductors CrSBr [8–11], NiPS₃ [?], MnPS₃ [12] or van der Waals (vdW) heterostructures [? ?] of CrI₃ 2D ferromagnetic semiconductor. Magnon propagation in these experiments involving the dynamics of localized magnetic moments on Cr atoms, which the LLG equation views as unit vectors of fixed length $\mathbf{M}_i$, has been successfully explained by the LLG equation. Its usage for Cr atoms hosting spins $S = 3/2$ can be justified from open quantum system theory, where the initial quantum dynamics of spins $S = 3/2$ transition into classical LLG dynamics in the long-time limit [13].

However, these experiments have also required additional damping features [11] justified by fitting it to the data. On the other hand, recent advances have made possible rigorous derivations of extended LLG equations, such as from open quantum system theory [13–17] or from Schwinger-Keldysh nonequilibrium quantum field theory (SKFT) [18? –22], demonstrating the possibility of a much richer structure of magnetization damping that sensitively depends on the microscopic details of the dissipative bath acting on LMMs $\mathbf{M}_i$. The baths considered thus far have included electrons [19? , 20], electrons driven by light [20] or phonons [18, 21, 22] or quantum magnons [?] which do not interact with each other within the bath.

The bath of excitons has not been considered yet, as this is a challenging problem due to interactions of particles within the bath. Excitons are quasiparticles composed of an electron-hole pair bound by Coulomb interaction [? ? ? ? ? ?]. Experimental imaging [? ?] shows their peculiar features, such as particularly large spatial extent, in the case of CrSBr [23? ? ?].

In this Letter, we derive a semiclassical equation for classical magnetic moments $\mathbf{M}_L(t)$ representing the macrospin localized in each layer ($L = 1, 2$) of an generic magnetic semiconductor. We find that as light-activated exciton hop between layers, effective torque and damping processes are induced in the classical moments, described by a generalized LLG equation [17, 19, 24] :

$$\partial_t \mathbf{M}_L = \mathbf{M}_L \times [\mathbf{B}_{cl}^L + j_\theta^L \mathbf{M}_L + \sum_{L'} \eta_\theta^{LL'} \partial_t \mathbf{M}_{L'}]. \quad (1)$$

The torque magnitude j_θ^L is found to increase with the number of excitons (or, complementary, with the fluence of the light pulse), which agrees with the available experimental observation [11, 25, 26]. Complementary, we find that the damping $\eta_\theta^{LL'}$ is not the conventional Gilbert [2] type, and does not vary with the number of excitons (also in agreement with the experimental observation). We also find that delocalized excitons mediate both of these processes with a greater efficiency than tightly bound counterparts. Overall, we provide a microscopically justified framework to describe the magnetic dynamics in the presence of excitons, resulting in an effective semiclassical equation [Eq. (1)] as our principal result.

Models and Methods—Let us consider the following microscopic quantum-classical Hamiltonian of a magnetic semiconducting bilayer

$$\hat{H} = \mathcal{H}_{cl} + \hat{H}_{ex} + \cos(\theta/2) \hat{X}, \quad (2)$$

where $\mathcal{H}_{cl}$ is the classical Hamiltonian describing the localized magnetic moments, note that we consider the macrospin approximation, effectively reducing the classical variables to one localized magnetic moment per layer [see End Matter for details on the classical Hamiltonian]. Furthermore, $\hat{H}_{ex}$ is the electronic Hamiltonian, θ is the angle between the two macrospins [i.e., $\cos(\theta) = \mathbf{M}_1 \cdot \mathbf{M}_2$] and $\hat{X}$ is an electronic operator, which we subsequently detail.

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The electronic Hamiltonian can be further decomposed as

$$\hat{H}_{\text{ex}} = \hat{H}_{\text{hop}} + \hat{H}_{\text{gap}} + \hat{H}_{\text{U}} + \hat{H}_{\text{CT}}, \quad (3)$$

where $\hat{H}_{\text{hop}}$ describes the intralayer hopping of electrons in both the valence and conduction bands, which we consider one-dimensional as justified by the relatively flat conduction bands observed in CrSBr [8, 27]

$$\hat{H}_{\text{hop}} = \sum_{\langle ij \rangle} \sum_{L=1}^2 \sum_{b \in \{c,v\}} t_b (\hat{c}_{i,L}^{b\dagger} \hat{c}_{j,L}^b + \text{H.c.}), \quad (4)$$

where $\hat{c}_{i,L}^b$ ($\hat{c}_{i,L}^{b\dagger}$) is the creation (annihilation) fermionic operator for electron in site i of layer L . Additionally, b super-index signals the band [here c and v indicates the conduction and valence bands, respectively], $t_c = -0.35$ eV, $t_v = 0.45$ eV, and $\langle ij \rangle$ indicates sum over intralayer nearest neighbors. Let us note that we assume the spin of electrons in each layer to be implicitly aligned with the corresponding macrospin. This is justified as in magnetic semiconductors, magnetic order and excitons originate from the same orbitals [25, 27, 28]. Therefore, a spinful electronic Hamiltonian would have spin minority bands at higher energies and its impact in the magnon-exciton coupling is negligible. Additionally, the term $\hat{H}_{\text{gap}}$ opens a gap between the conduction and valence band

$$\hat{H}_{\text{split}} = \Delta/2 \sum_i \sum_{L=1}^2 (\hat{c}_{i,L}^{c\dagger} \hat{c}_{i,L}^c - \hat{c}_{i,L}^{v\dagger} \hat{c}_{i,L}^v). \quad (5)$$

Here we set $\Delta/2 = 3.6$ eV, which set the gap at 2.1 eV. Furthermore, we include intralayer Coulomb binding interaction

$$\hat{H}_{\text{Coulomb}} = U \sum_i \sum_{L=1}^2 \hat{n}_{i,L}^c \hat{n}_{i,L}^v, \quad (6)$$

where $\hat{n}_{i,L}^b = \hat{c}_{i,L}^{b\dagger} \hat{c}_{i,L}^b$ is the number operator, and U is left as a free parameter as it ultimately determines the localization of the exciton. Additionally, we consider an interlayer Coulomb interaction

$$\hat{H}_{\text{CT}} = V \sum_i (\hat{n}_{i,1}^c + \hat{n}_{i,1}^v) \cdot (\hat{n}_{i,2}^c + \hat{n}_{i,2}^v), \quad (7)$$

where V is also set as a free parameter, this interlayer coupling has been suggested in previous theoretical works [29] focusing on static properties of CrSBr. Nevertheless, its impact on the exciton-mediated torque on magnon is unknown. Finally, the operator $\hat{X}$ connecting to the classical θ variable is the interlayer hopping

$$\hat{X} = \sum_{b \in \{c,v\}} \sum_i t_b^{\text{inter}} (\hat{c}_{i,1}^{b\dagger} \hat{c}_{i,2}^b + \text{H.c.}), \quad (8)$$

Here, $t_c^{\text{inter}} = 0.064$ eV and $t_v^{\text{inter}} = 0.029$ eV. These parameters determine the exciton binding energy renormalization observed in the presence of an external magnetic

field [27, 29]. Note that this quantum-classical coupling, first proposed by Anderson in Ref. [30], dress the interlayer hopping according to its implicit spin orientation [forbidding hopping if the moments are counteraligned].

A. Spin and Exciton response

I. END MATTER

A. Quantum Action

Starting from the analogue quantum model of Eq (2), and using the standard $SU(2)$ coherent states $\{|\mathbf{M}\rangle = |\mathbf{M}_1\rangle|\mathbf{M}_2\rangle\}$ for the spins and fermionic coherent states $\hat{c}_{i,L}^b|c_{i,L}^b\rangle = c_{i,L}^b|c_{i,L}^b\rangle$ for the electrons we get the total action $\mathcal{S} = \mathcal{S}_{cl} + \mathcal{S}_e$

$$\mathcal{S} = \quad (9)$$

where the self-energy Σ should be computed including coherent or incoherent exciton effects [31, 32]. In the manuscript we compute the total $\hat{\rho} = -iG^<(t, t)$ and $G^<(t, t')$ using DMRG. Additionally, we pick the HSEX approximation for the selfenergy in order to calculate the where $\mathcal{H}_{cl} \equiv \langle \mathbf{M} | \hat{H}_{cl} | \mathbf{M} \rangle$

B. Semiclassical limit and adiabatic expansion

C. Effective exciton action

In order to capture excitonic effects,

$$\mathcal{S}_{\Phi}^{\text{eff}} = \int_{\mathcal{C}} dz \sum_{\mathbf{Q}, \lambda} \bar{\Phi}_{\mathbf{Q}}^{\lambda} [\delta_{\lambda\lambda'} K_{\lambda}^{-1} - \Pi_{\mathbf{Q}}^{\lambda\lambda'}] \Phi_{\mathbf{Q}}^{\lambda} \quad (10)$$

The bubble-diagram becomes

$$\Pi^{\lambda\lambda'} = -i\text{Tr}[G_c \Gamma^{\lambda} G_v \Gamma^{\lambda'\dagger}] \quad (11)$$

$$(12)$$

we use the basis that diagonalizes $\hat{H}_0$ to compute the bubble diagram

$$V^{\lambda\lambda'} \doteq \begin{bmatrix} \bar{V} & \Delta V & 0 & 0 \\ \Delta V & \bar{V} & 0 & 0 \\ 0 & 0 & \bar{V} & \Delta V \\ 0 & 0 & \Delta V & \bar{V} \end{bmatrix} \quad (13)$$

$\bar{V} = \frac{1}{2}(V_{\text{IN}} + V_{\text{CT}})$ and $\Delta V = \frac{1}{2}(V_{\text{IN}} - V_{\text{CT}})$ and the bubble elements becomes diagonal in that basis

$$\Pi^{\lambda\lambda'} \doteq -i \begin{bmatrix} G_c^+ G_v^+ & 0 & 0 & 0 \\ 0 & G_c^- G_v^- & 0 & 0 \\ 0 & 0 & G_c^+ G_v^- & 0 \\ 0 & 0 & 0 & G_c^- G_v^+ \end{bmatrix} \quad (14)$$

Using Langreth rules and the explicit expression of the Green functions we get:

The retarded component of the Bubble diagram becomes

$$\Pi_Q^{R, \lambda\lambda'}(\Omega) = \frac{1}{N_k} \sum_k \frac{f(\epsilon_{c, k+Q}^{\lambda}) - f(\epsilon_{vk}^{\lambda'})}{\Omega - (\epsilon_{c, k+Q}^{\lambda} - \epsilon_{vk}^{\lambda'}) + i\eta} \quad (15)$$

The eigen energies of $\hat{H}_0$

$$\epsilon_{ak}^{\pm} = \epsilon_{ak} \pm \gamma_a \cos(\theta/2) \quad (16)$$

The energy of the exciton is determined by the poles of the retarded component of the dressed interaction. Note it shares poles with the exciton susceptibility $\det[\Pi_{Q=0}^{-1}(\Omega) - V] = 0$, where we can identify the BSE Hamiltonian for the exciton $\hat{H}_{\Phi} - \omega = \hat{\Pi}^{-1}(\omega) - \hat{V}$

With the given angular dependence of the Hamiltonian in Eq. [], the effective spin operator can be obtained varying the complete action with respect to the magnetization $\frac{\delta \mathcal{S}_e}{\delta \mathbf{M}} = \mathbf{B}_e$

$$\mathbf{B}_e(t) = i\text{Tr}[\hat{G}^<(t) \hat{X}] \quad (17)$$

In order to separate the exciton torque, we pick the exciton channel to decouple the action, and get an effective theory around exciton field Φ

D. Exciton torque and damping

The dressed exciton propagator D and the exciton spin susceptibility χ_{Φ} are related through

$$\chi_{\Phi} = \Pi + \Pi \circ D \circ \Pi, \quad (18)$$

The exciton susceptibility is approximately computed assuming the presence of quasi-stationary exciton and vanishing polarization [31–33]. The retarded and lesser components of the exciton susceptibility reads

$$\chi_{\mathbf{Q}}^R(\omega) = i \sum_n Y_{\mathbf{Q}}^n \frac{s_n}{\omega - \Omega_{\mathbf{Q}}^n} \bar{Y}_{\mathbf{Q}}^n. \quad (19)$$

Additionally in the regime of low density excitations, the lesser component can be approximated by

$$\chi_{\mathbf{Q}}^<(\omega) = 2\pi \sum_n F_{\mathbf{Q}}^n Y_{\mathbf{Q}}^n \delta(\omega - \Omega_{\mathbf{Q}}^n) \bar{Y}_{\mathbf{Q}}^n \quad (20)$$

$$F_{\mathbf{Q}}^n = \sum_{bb'\mathbf{Q}'} \bar{Y}_{b\mathbf{Q}'}^n \frac{f_b \bar{f}_{b'}}{(f_{bb'\mathbf{Q}'}^{\mathbf{Q}})^2} Y_{b'\mathbf{Q}'} \quad (21)$$

In order to calculate the spin torque and damping induced by the (un)bound electron-hole pairs, we use an adiabatic assumption [16, 17], justified by the fast electron-hole degrees of freedom in comparison with the magnetization dynamics [8]. Considering the effective action in [Eq. (??)], we derive an expression for the torque j_{θ}^L and damping magnitude $\eta_{\theta}^{LL'}$

E. Classical magnetic Hamiltonian

The classical Hamiltonian, entering into the LLG Eq. (1), is given by

$$\mathcal{H}(t) = J^{\text{AF}} \mathbf{M}_1 \cdot \mathbf{M}_2 - g\mu_B B \sum_{L=1}^2 \mathbf{M}_L \cdot \mathbf{e}_y + \sum_{L=1}^2 \left[K_z (\mathbf{M}_L \cdot \mathbf{e}_z)^2 - K_x (\mathbf{M}_L \cdot \mathbf{e}_x)^2 \right] \quad (22)$$

Such an effective classical Heisenberg Hamiltonian of LMMs can be extracted from first-principles calculations [34–36] or by fitting magnon spectra from neutron scattering data [37]. Here $J^{\text{AF}} = 0.0195$ eV is the FM exchange coupling; $K_x = 0.021$ eV and $K_z = 0.057$ eV specify the magnetic anisotropy along the x - and z -axis, respectively; $g\mu_B B = 0.042$ eV is the external magnetic field applied along the y direction, which induces the non-collinearity of the LMMs; for the canting external magnetic field along the y direction, we use $g\mu_B B = 0.042$ eV; Note that J^{AF} , K_x , K_z , B parameters were also employed to interpret experiments [8–10].

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